

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:	03/08/2026	Duration:	60 minutes

Code of Conduct

I Ch.B.Vinay (192425148) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *ch.B.Vinay*

Assigned Scenario: Intelligent Supply Chain Risk Prediction and Vendor Management Platform

1. Study and Analysis of the Scenario

1.1 Problem Statement

Modern enterprises depend on complex, multi-tier global supply chains spanning raw-material suppliers, manufacturers, logistics providers, and distributors. Disruptions caused by geopolitical instability, natural disasters, vendor financial distress, and logistics delays can halt production and inflate costs before organizations even become aware of the risk.

Traditional vendor management relies on periodic manual reviews and reactive escalation once a disruption has already occurred, offering little advance warning. The proposed Intelligent Supply Chain Risk Prediction and Vendor Management Platform ingests data from enterprise resource planning (ERP) systems, logistics/shipping trackers, market and news feeds, and vendor financial rating services to continuously predict disruption risk, score and rank vendors, recommend alternate sourcing options, and automate compliance monitoring.

1.2 Business Domain

Supply chain management and procurement operations - specifically AI-driven vendor risk analytics, disruption prediction, and sourcing optimization for manufacturing, retail, and distribution enterprises with multi-region vendor networks.

1.3 Stakeholders

- Supply Chain Manager - monitors vendor risk, oversees disruption response, and approves alternate sourcing decisions.
- Procurement Officer - searches, compares, and onboards vendors; manages contracts and purchase orders.
- Risk Analyst - configures risk-scoring models, investigates flagged vendors, and validates prediction accuracy.
- System Administrator - configures data source integrations, risk thresholds, and manages user accounts.
- Executive Management / CFO - uses risk and spend analytics for strategic sourcing and continuity planning.
- Vendor / Supplier - external party providing goods/services and compliance documentation.

- ERP System, Logistics/Shipping Tracking System, News/Market Data Feed, and Vendor Financial Rating Service - external systems that supply or consume data.

1.4 Existing Challenges

- Vendor risk assessment is largely manual, periodic, and reactive rather than continuous.
- Disruptions (port closures, factory shutdowns, financial distress) are often detected only after delivery delays occur.
- Vendor performance, financial, and logistics data are siloed across ERP, procurement, and shipping systems.
- No predictive insight into how geopolitical, weather, or economic events will affect specific vendors or regions.
- Difficult to objectively compare and rank vendors or identify qualified backup suppliers during a crisis.

1.5 System Objectives

- Monitor vendor risk in real time across financial, geopolitical, operational, and compliance dimensions.
- Predict the likelihood and severity of supply chain disruption per vendor and region.
- Score and rank vendors based on predicted risk and historical performance.
- Recommend qualified alternate/backup vendors when risk exceeds acceptable thresholds.
- Automate vendor onboarding and ongoing compliance checks.
- Generate risk exposure and vendor performance analytics for procurement and executive stakeholders

2. Functional and Non-Functional Requirements

2.1 Functional Requirements

ID	Requirement	Priority
FR-01	The system shall collect vendor master, transaction, and purchase-order data from ERP systems.	High
FR-02	The system shall integrate with logistics/shipping tracking systems to capture shipment status and delay data.	High

ID	Requirement	Priority
FR-03	The system shall integrate with news and market data feeds to capture geopolitical, weather, and economic risk signals.	High
FR-04	The system shall predict disruption risk per vendor/region using historical and real-time data (ML-based forecasting).	High
FR-05	The system shall recommend alternate/backup vendors based on predicted risk and performance.	High
FR-06	The system shall allow supply chain managers to view real-time and historical vendor risk dashboards.	High
FR-07	The system shall automatically flag vendors exceeding configured risk thresholds for review.	Medium
FR-08	The system shall generate vendor performance and risk-exposure reports for management.	Medium
FR-09	The system shall send alerts for high-risk events, delayed shipments, or vendor compliance issues.	Medium
FR-10	The system shall allow procurement officers to search, compare, and onboard vendors based on predicted risk.	Medium
FR-11	The system shall allow administrators to configure data sources, risk thresholds, and scoring weights.	High
FR-12	The system shall maintain an audit log of risk scores, alerts, and procurement decisions.	Medium

2.2 Non-Functional Requirements

Category	Requirement
Performance	The system shall ingest vendor, logistics, and market data streams and refresh the risk dashboard within 5 seconds under normal load.
Security	All vendor, financial, and logistics data shall be encrypted in transit (TLS 1.2+) and at rest; access shall be role-based (procurement officer, supply chain manager, risk analyst, admin).
Reliability	The system shall maintain 99.5% uptime and gracefully handle data-source outages via cached last-known risk scores.

Category	Requirement
Usability	The dashboard shall be usable by procurement staff without specialized technical training and shall follow accessible design (WCAG 2.1 AA).
Scalability	The architecture shall scale from a single procurement category pilot to an enterprise-wide, multi-region vendor network with thousands of vendors.
Maintainability	The system shall be modular (separate ingestion, risk-scoring, recommendation, and reporting services) to allow independent updates.
Availability	Core monitoring and dashboard functionality shall be available 24/7, excluding scheduled maintenance windows.
Interoperability	The system shall integrate via open REST APIs with ERP systems, logistics tracking platforms, financial rating services, and market/news data feeds.

3. Actors and Use Cases

3.1 Actors

Actor	Type	Key Use Cases
Supply Chain Manager	Primary	Monitor Vendor Risk, Predict Disruption Trends, View Risk Analytics Dashboard, Receive Risk Alerts, Approve Alternate Vendor, Generate Risk & Performance Reports
Procurement Officer	Primary	Search/Compare Vendors, Onboard Vendor, View Risk Analytics Dashboard
Risk Analyst	Primary	Configure Risk Models, Review Flagged Vendors, Generate Risk & Performance Reports
System Administrator	Primary	Configure Data Sources & Thresholds, Manage User Accounts, Generate Reports
Vendor / Supplier	Secondary (External System)	Provide compliance and performance data during onboarding and audits

Actor	Type	Key Use Cases
ERP System	Secondary (External System)	Feed vendor transaction and purchase-order data into the ingestion pipeline
Logistics / Shipping Tracking System	Secondary (External System)	Provide shipment status and delay data for risk scoring
News / Market Data Feed	Secondary (External System)	Supply geopolitical, weather, and economic risk signals
Vendor Financial Rating Service	Secondary (External System)	Provide vendor creditworthiness and financial-stability data

3.2 Use Case Descriptions

Use Case	Actor(s)	Description
Monitor Vendor Risk	Supply Chain Manager	System displays live, per-vendor and per-region risk scores derived from ERP, logistics, market, and financial data.
Predict Disruption Trends	Supply Chain Manager, Risk Analyst	System forecasts likelihood and severity of disruption per vendor/region using historical and real-time data.
Recommend Alternate Vendors	Supply Chain Manager	System suggests qualified backup vendors when a primary vendor's risk exceeds acceptable thresholds.
Search / Compare Vendors	Procurement Officer	Officer searches and compares vendors by risk score, performance, category, and region.
Onboard Vendor	Procurement Officer, Vendor/Supplier	Officer registers a new vendor; system runs automated compliance and risk checks before activation.

Use Case	Actor(s)	Description
View Risk Analytics Dashboard	Supply Chain Manager, Procurement Officer, Risk Analyst	Users view real-time and historical vendor risk and performance visualizations.
Receive Risk Alerts	Supply Chain Manager	System notifies the manager of high-risk events, shipment delays, or compliance issues.
Approve Alternate Vendor	Supply Chain Manager	Manager reviews and approves a recommended backup vendor during a disruption event.
Review Flagged Vendors	Risk Analyst	Analyst investigates vendors flagged for exceeding configured risk thresholds.
Configure Risk Models	Risk Analyst	Analyst tunes risk-scoring models, weights, and prediction parameters.
Generate Risk & Performance Reports	Supply Chain Manager, Risk Analyst, System Administrator	System compiles vendor risk-exposure and performance reports for review.
Configure Data Sources & Thresholds	System Administrator	Admin registers integration endpoints and sets risk thresholds used for alerts and flagging.
Manage User Accounts	System Administrator	Admin creates, updates, and deactivates user accounts and role permissions.
Ingest ERP & Logistics Data	ERP System, Logistics/Shipping Tracking System	System continuously ingests and normalizes vendor transaction and shipment data.
Ingest Market / News Risk Signals	News/Market Data Feed	System ingests geopolitical, weather, and economic event data relevant to vendor regions.
Sync Financial Rating Data	Vendor Financial Rating Service	System pulls updated creditworthiness and financial-stability ratings for vendors.

3.3 Use Case Diagram

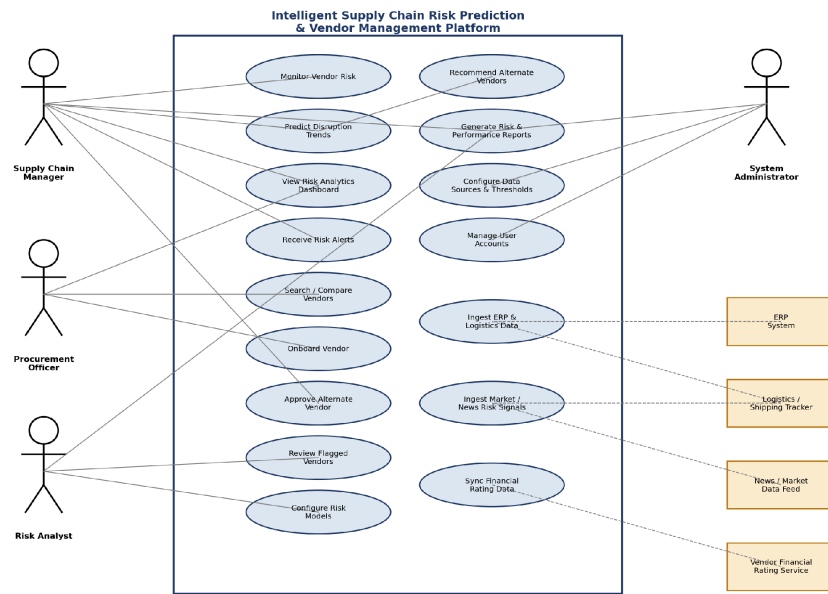


Figure 1: Use Case Diagram - Intelligent Supply Chain Risk Prediction and Vendor Management Platform

4. Software Requirement Specification (SRS)

4.1 Introduction

This SRS describes the requirements for an Intelligent Supply Chain Risk Prediction and Vendor Management Platform intended to collect data from ERP systems, logistics/shipping trackers, market and news feeds, and vendor financial rating services to predict disruption risk, score and rank vendors, and support proactive sourcing decisions across a multi-region vendor network.

4.2 Problem Description

Refer to Section 1.1 for the full problem statement. In summary, the system replaces periodic, manual vendor risk reviews with a unified, AI-driven platform that continuously predicts disruption risk, recommends alternate vendors, and gives procurement and executive stakeholders actionable analytics to reduce supply chain exposure and cost.

4.3 Scope of the System

In scope: real-time vendor risk monitoring, disruption prediction, vendor scoring and ranking, alternate-vendor recommendation, vendor search and onboarding, risk/performance dashboards and reports, data-source and threshold configuration, and automated alerting.

Out of scope (Phase 1): automated contract negotiation and execution, full financial settlement/payment processing, integration with in-transit IoT cargo sensors, and multi-tier (sub-supplier) risk mapping beyond direct vendors (planned for a later phase after single-category pilot validation).

4.4 Functional Requirements

See Section 2.1 for the complete list (FR-01 to FR-12).

4.5 Non-Functional Requirements

See Section 2.2 for the complete list (performance, security, reliability, usability, scalability, maintainability, availability, interoperability).

4.6 Use Case Descriptions

See Section 3.2 and the diagram in Section 3.3.

4.7 Data Requirements

- Vendor data: vendor ID, name, category, region, contract details, compliance status.
- ERP data: purchase order ID, vendor ID, order amount, delivery date, payment terms.
- Logistics data: shipment ID, vendor ID, origin/destination, status, expected/actual delivery date.
- Market/risk event data: event type, region, severity, timestamp, source.
- Financial rating data: vendor ID, credit score, financial-stability index, rating date.
- Prediction data: vendor ID, predicted risk score, risk category, prediction window, model confidence score.
- User account data: role (procurement officer/supply chain manager/risk analyst/admin), credentials (hashed), access permissions.

4.8 External Interface Requirements

- User Interface: web-based risk analytics dashboard for supply chain managers, risk analysts, and administrators; procurement portal for vendor search, comparison, and onboarding.

- Hardware Interface: none mandatory (cloud-based); optional IoT integration for in-transit cargo condition sensors in later phases.
- Software Interface: integration APIs with ERP systems (e.g., SAP/Oracle), logistics/shipping tracking platforms, vendor financial rating services, and market/news data providers.
- Communication Interface: HTTPS/REST APIs for dashboard, vendor onboarding, and third-party integrations; webhook subscriptions for near-real-time risk event feeds.

4.9 System Features

- Real-time vendor risk dashboard with per-vendor/region scores and historical trends.
- AI-based disruption prediction engine using historical and streaming data.
- Automated alternate-vendor recommendation engine based on predicted risk and performance.
- Configurable alerts for high-risk events, shipment delays, and compliance issues.
- Automated vendor onboarding compliance checks.
- Risk exposure and vendor performance analytics reports for procurement and executive stakeholders.

5. Assumptions and Constraints

5.1 Assumptions

- ERP, logistics tracking, financial rating, and market/news data APIs are available, documented, and accessible.
- Sufficient historical vendor, shipment, and risk-event data is available, or will be collected during a pilot period, to train the prediction model.
- Vendors provide accurate compliance and performance data during onboarding and periodic reviews.
- Procurement and supply chain staff will act on system recommendations and alerts in a timely manner.
- Network connectivity is reliable enough to support near-real-time ingestion of external data feeds.

5.2 Constraints

- Technical: prediction accuracy depends on the quality and availability of third-party data feeds; integration with legacy ERP systems may require custom adapters.
- Operational: risk-scoring models require periodic retraining and calibration; procurement staff need training to interpret dashboards and act on recommendations.
- Economic: costs of market/financial data subscriptions and cloud analytics infrastructure must fit within the organization's allocated budget.
- Legal/Compliance: the system must comply with data-privacy, vendor-confidentiality, and applicable export-control regulations when processing geopolitical and financial data.
- Time: initial deployment is expected as a single-category vendor pilot before full enterprise-wide rollout, within the agreed project timeline.

6. Validation of Requirement Completeness and Consistency

6.1 Stakeholder Coverage Check

Each identified stakeholder (Section 1.3) has at least one corresponding functional requirement and use case: supply chain managers (FR-04 to FR-06, FR-09, FR-12), procurement officers (FR-10), risk analysts (FR-04, FR-07, FR-08), system administrators (FR-11, FR-12), and executive management (FR-08). External systems - ERP, logistics tracking, market/news feed, and financial rating service - are covered by FR-01, FR-02, FR-03, and FR-03 respectively. This confirms all stakeholder needs identified during scenario analysis, including all six stated objectives, are addressed.

6.2 Ambiguity, Redundancy, and Inconsistency Check

- Ambiguity: requirement wording was reviewed to avoid vague terms (e.g., 'fast', 'reliable') without measurable criteria; NFRs were rewritten with specific, testable targets (e.g., '5 seconds', '99.5% uptime', 'thousands of vendors').
- Redundancy: 'Generate Risk & Performance Reports' is shared between the supply chain manager, risk analyst, and system administrator by design (each generates reports for a different audience/purpose) - this overlap is intentional and documented in Section 3.1/3.2, not a duplicate requirement.
- Inconsistency: cross-checked that every functional requirement maps to at least one actor/external system and use case, and that no use case exists without a supporting functional requirement.

- Missing requirements identified during review and added: audit logging (FR-12) and interoperability NFR (REST API integration with financial rating and market data services), both initially overlooked in the first draft.

6.3 Suggested Improvements

- Add a formal requirement traceability matrix (RTM) linking each FR/NFR to its originating objective/expected outcome and test case.
- Define explicit acceptance criteria (e.g., risk-prediction accuracy thresholds, alert response time) for high-priority requirements such as disruption prediction and vendor flagging.
- Conduct a stakeholder walkthrough with procurement and risk teams to validate risk-scoring weights before full deployment.
- Consider a future phase for multi-tier (sub-supplier) risk mapping and integration with contract negotiation/execution systems once the single-category pilot is validated.